Haobin Chen

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EDUCATION

South China University of Technology (SCUT) Guangzhou, China
Master of Engineering in Mechanical Engineering *Sep. 2024 – Jun. 2027*
Admitted through Recommendation *(Exempt from the National Postgraduate Entrance Examination).*

- **GPA:** 85.7/100;
- **Awards**: Postgraduate Scholarships 2024, 2025 & 2026

Southwest Petroleum University (SWPU) Chengdu, China
Bachelor of Engineering in Mechanical Design, Manufacturing and Automation *Sep. 2020 – Jun. 2024*

- **GPA:** 88.7/100; Overall Ranking: 2/94
- **Awards**: Wewodon Scholarship (1%, 2022); First-Class Scholarship (3%, 2022 & 2023)
- Undergraduate thesis: ranked 1st in defense; University Thesis Innovation Award (sole recipient across the major)

PUBLICATIONS & PATENTS

- **H. Chen**, Y. Chen, L. Xie, "PMG-RL for Stable Multi-Speed Walking Control of a Human Musculoskeletal Model," *2026 IEEE 3rd International Conference on Energy and Electrical Engineering*, Jun. 2026. (Accepted & Indexed by EI)
- **H. Chen**, F. Li, L. Xie, et al., "NORM-FoG: Normal-Gait-Oriented Residual Modeling for IMU-Based Freezing-of-Gait Detection," *Expert Systems with Applications*. (Under Review)
- L. Chen, **H. Chen**, L. Xie, et al., "FrontalSleepNet: Multiscale Context-Aware Sleep Staging from Information-Limited Frontal EEG," *IEEE Journal of Biomedical and Health Informatics*. (Under Review)
- **H. Chen**, Y. Chen, L. Xie, et al., "CPG-Guided Reinforcement Learning for Hip Assistance Strategy in a Musculoskeletal Environment," *IEEE Robotics and Automation Letters*. (In Preparation)
- H. Wang, **H. Chen**, et al., "*A Device and Method for Testing Phase-Change Thermal Storage Material Performance*," China Invention Patent ZL202311331955.0, Granted Oct. 2025.
- H. Wang, **H. Chen**, et al., "*A Phase-Change Material Thermal Storage Performance Testing Device,*" China Utility Model Patent ZL202322761347.5, Granted May 2024.
- Y. Wang, Y. Mao, **H. Chen**, et al., "*A Novel Infrared Non-Destructive Fruit Sugar Content Measurement Devic*e," China Utility Model Patent ZL202223027103.6, Granted Jun. 2023.
- H. Wang, H. Shi, **H. Chen**, et al., "*A Machine-Vision-Based Component Sorting Device*," China Utility Model Patent ZL202120585669.7, Granted Nov. 2021.

RESEARCH EXPERIENCE

Adaptive Hip Exoskeleton System for Freezing-of-Gait in Parkinson's Disease Guangzhou, China
Guangdong S&T Program (Grant No. 2026B0101110003)
Advisor: Dr. Longhan Xie *Jul. 2025 – Present*

- Developed a portable hip exoskeleton delivering 17 Nm peak assist torque with a 100 Hz real-time control loop for freezing-of-gait assistance in Parkinson's disease;
- Proposed NORM-FoG, a residual-modeling detector scoring deviations from subject-specific normal gait, achieving 72.2% average precision and outperforming the strongest evaluated baseline by 3.1 percentage points;
- Built a multi-location IMU dataset from 12 patients with PD (152.56 min of annotated gait data) and conducted within-subject and leave-one-subject-out validation;
- Manuscript under review at Expert Systems with Applications; companion control work (CPG-guided RL for hip assistance, tested in a MuJoCo musculoskeletal environment) targeting IEEE Robotics and Automation Letters.

EEG-fNIRS Multimodal Brain-Controlled Rehabilitation Robot Guangzhou, China
Natural Science Foundation of Guangdong Province (Grant No. 2024A1515011190)
Advisor: Dr. Longhan Xie *Sep. 2024 – Jun. 2025*

- Developed a low-channel prefrontal EEG-fNIRS acquisition framework replacing full-scalp caps for brain-controlled rehabilitation, reducing setup time from over 15 min to 3 min;

- Suppressed cross-modal crosstalk through spatially isolated FPC layouts and time-division-multiplexed illumination; differential front-ends achieved input noise < 3 μV and time-constant error < 2%;
- Contributed to system commercialization; the device obtained medical-device registration in China, supporting its transition from a laboratory prototype to clinical deployment.

Dual-Manipulator Cooperative System for Drilling Derrick Pin Assembly and Disassembly Chengdu, China
PetroChina Project
Advisor: Teng Hu *Sep. 2023 – Dec. 2024*
- Designed a force–position hybrid cooperative architecture coordinating two collaborative robots (alignment/insertion) with one hydraulic arm (reaction-force support) for automated pin handling under high reaction loads;
- Engineered a gear-transmission end-effector with a closed structural force-transfer path to offload robot-end loads; a payload–workspace trade-off analysis drove robot selection and station layout.

Phase-Change Material Property Testing Instrument Design Chengdu, China
Advisor: Haohan Wang *Apr. 2023 – Jun. 2024*
- Built a low-cost PCM performance tester (< ¥100/test, ~45% lower than DSC/TGA testing) via latent-heat FEA modeling; 5-step workflow enabling high-throughput thermal-storage evaluation.

Apple Sugar Content Distribution Monitoring and Management System Chengdu, China
Advisor: Yunxu Zhou *Jun. 2022 – Oct. 2023*
- Developed an NIR-based nondestructive sugar-content measurement device, achieving an R^2 of 0.968 for sugar-content prediction; the portable device can be deployed for apple quality monitoring and classification.

PROFESSIONAL EXPERIENCE

Lychee Intelligent Technology Co., Ltd. Guangzhou, China
R&D Engineer *Jun. 2025 – Dec. 2025*
- Commercialized the SCUT hip exoskeleton through 5+ design iterations, achieving small-batch production of 100+ units and reducing unit cost by nearly 75%;
- Led the regulatory pipeline, securing China Compulsory Certification and medical-device registration, completing the transition from prototype to engineered product.

Lincseek Technology Co., Ltd. Chengdu, China
R&D Engineer *Nov. 2023 – Jun. 2024*
- Built a six-DOF serial robot prototype; independently developed forward/inverse kinematics algorithms, motion-control software, and a PC-based graphical user interface.

TEACHING

Engineering Training Center, Southwest Petroleum University Chengdu, China
Teaching Assistant *Sep. 2021 – Jun. 2023*
- Assisted lab instruction for the Engineering Cognition Practice course; supported robotics-lab management and competition mentoring for 30+ students.

HONORS

- National Gold Award, China International College Students' Innovation Competition (2025)
- Provincial Grand Prize, 18th Guangdong Extracurricular Academic Science and Technology Competition (2025)
- National First Prize, 16th National Undergraduate Advanced Mapping Technology and Product Information Modeling Innovation Competition (Additive Manufacturing Track, 2023)
- National First Prize, National Intelligent Robot Creative Design Competition for University Students (2023)

TECHNICAL SKILLS

- **Robotics & Simulation:** FK/IK & dynamics, motion planning, ROS, MuJoCo (sim-to-real, RL integration)
- **Hardware & EDA:** SolidWorks, UG, Ansys (CAD/CAE/CAM); EasyEDA
- **Programming:** Python, C, C++, MATLAB
- **English**: IELTS 7.0 (L6.5, R8.0, W7.0, S6.0)